MADISON VAN HORN MITCHELL

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WORK EXPERIENCE

CALTECH VISION LAB| Pasadena, CA 07/2024 – Present

Lead engineer for an edge-deployed computer vision system for salmon detection, tracking, and counting in low signal-to-noise data.

Machine Learning Engineer

- Built and maintained production ML infrastructure for data collection and ingestion, model training, evaluation, benchmarking, and deployment.
- Reduced inference latency by 2x through preprocessing optimization and hyperparameter tuning.
- Deployed an offline computer vision application for population counting on NVIDIA Jetson devices during the Klamath River dam removals, one of the largest ecological restoration efforts in U.S. history.
- Extended the application to macOS, Windows and AWS EC2, enabling both local and cloud-based workflows for researchers and field teams.
- Managed data operations across 12TB of imaging data including collection, annotation, and quality control for model training and evaluation.
- Designed and implemented a PostgreSQL metadata database with SQLAlchemy, replacing manual tracking of sonar imagery and deployment metadata with a queryable system for researchers.
- Partnered with research scientists and field stakeholders to translate operational requirements into production ML system capabilities.
- Co-authored a paper under review extending a large-scale sonar imaging dataset with 1M+ bounding boxes, 2x as many videos as the original version, along with new evaluation protocols and metrics for out-of-distribution data.

MANTIUM | Remote, U.S. 02/2022 – 02/2024

A cloud-based AI platform that raised $12.75M in seed funding for building with large language models and managing them at scale.

NLP Engineer

- Developed Retrieval-Augmented Generation (RAG) pipelines for document QA systems, integrating Redis-backed vector retrieval, and citation-grounded conversational responses.
- Built an evaluation framework for benchmarking sparse and dense retrievers across retrieval accuracy, indexing speed, and latency tradeoffs.
- Improved few-shot text classification performance by 9% on a large frozen language model by implementing contextual calibration to correct prompt-induced label bias.
- Fine-tuned text and multimodal models via supervised fine-tuning (SFT) for multi-label classification and question answering tasks.

NORTHROP GRUMMAN CORPORATION | San Diego, CA 06/2019 – 09/2020

Associate Software Engineer

- Maintained and integrated software for unmanned aircraft subsystems, including workstation configuration, software releases, and hardware testing workflows for system validation and issue resolution.

Software Engineer Intern and Software Quality Assurance Engineer Intern 05/2017 – 08/2018

- Supported software verification and validation efforts for unmanned aircraft systems through code reviews, standards compliance checks, and synthetic vision evaluation of safety-critical flight scenarios.

EDUCATION

MS, Artificial Intelligence | University of Edinburgh, U.K.

- Awarded Distinction.
- Dissertation: "Understanding How Immune Systems See Disease: Exploring Self-Attention of MHC Class I Presented Sequences with Natural Language Processing Techniques."

BS, Computer Science | San Diego State University, U.S.

- GPA: 3.69

PROFESSIONAL ACTIVITIES

- Co-organizer, 13th Workshop on Fine-Grained Visual categorization (FGVC). CVPR 2026. [page]
- Train and Deploy Large Language Models on Amazon SageMaker. AWS re:Invent. (November, 2022). [slides]

PUBLICATIONS

- Van Horn, M., Barry, D., Ranade, S., Van Horn, G., "Few-Shot Fine-Grained Visual Classification using Coarse-Grained Supervision," 8th Workshop on Fine-Grained Visual Categorization (FGVC8), IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), Nashville, TN, June 2021.
- Van Horn, M. "Using Deep Learning for Visual Decoding and Reconstruction from Brain Activity: A Review." arXiv preprint arXiv:2108.04169, 2021.

AWARDS AND RECOGNITION

- Placed top-6 out of 101 projects in the IBM Machine Learning Project Competition. Project title: "Few-Shot Fine-Grained Visual Classification using Coarse-Grained Supervision."
- Recognized as an Excellent performer in Northrop Grumman's 2019 yearly review.
- Received a Northrop Grumman Bravo Award for efforts in testing new hardware.
- Earned a Northrop Grumman Bravo eRecognition for final summer presentation.
- Graduated Magna Cum Laude from San Diego State University.
- Earned Dean's List five semesters at San Diego State University.
- Received Women's Club Scholarship three times for academics and community service.
- Received Wonderful Company Scholarship for academic excellence.

SELECTED TECHNICAL SKILLS

- Programming: Python
- Infrastructure & Cloud: AWS (EC2, S3, RDS, SageMaker), Docker, Linux, Slurm, Redis, NVIDIA Jetson
- ML & AI: PyTorch, Hugging Face Transformers, scikit-learn, LangChain, Haystack, NumPy, PIL, Pandas
- MLOps: Weights & Biases, TensorBoard
- Tools: Git, Jupyter, Matplotlib
- Areas of Interest: Image and video understanding, human-in-the-loop machine learning systems, applied machine learning to biodiversity monitoring and conservation problems, natural language processing, bioinformatics.

PROJECTS

- Analyzing and Improving the Performance of a Convolutional Neural Network with Vanishing Gradient Problem.
- Identifying and Investigating the Issue of Overfitting in a Deep Neural Network.
- Investigating Inhibitory Stabilized Networks, Paradoxical Inhibition, and the Stabilized Supralinear Network.
- Computational Modeling of Behavioral Data.
- Comparing Sensitivities of Classical and Deep Learning Methods Applied to Image Classification.